SOKOINE UNIVERSITY OF AGRICULTURE

Department of Informatics and Information Technology (DIIT)

BIT 111: MICROCOMPUTERS AND APPLICATIONS

TOPIC 5: MICROCOMPUTER APPLICATIONS — LECTURE COMPENDIUM

Audience: First-year undergraduate students
Approach: Introductory concepts, guided examples, and discipline-relevant applications
Topic overview: Word processing, spreadsheets, web use, and email

Topic 5: Lecture Outline & Learning Roadmap

Productivity Applications, Spreadsheets & Web Systems (Coverage: 5.1, 5.2, 5.3)

- • Section 5.1: Word Processing Systems & Academic Document Engineering
- - Structural document architecture: Character/paragraph formatting, section breaks, margins
- - Advanced automation: Heading styles, automated Table of Contents (TOC), cross-references
- - Mail merge architecture: Data sources, merge fields, and conditional rule filters (`IF...THEN`)
- • Section 5.2: Spreadsheet Applications & Simple Calculations
- - Worksheet layout: rows, columns, cells, and cell references
- - Simple arithmetic formulas and introductory `SUM` and `AVERAGE` functions
- - Organizing information with clear labels, basic sorting, and charts
- • Section 5.3: Web Browsing, Internet Protocols & Electronic Mail Architecture
- - World Wide Web: URLs, HTTP/HTTPS request-response cycles, status codes, and TLS certificates
- - Email protocols: SMTP (message submission/relay), POP3 vs IMAP4 (retrieval and sync)
- - Email security & anti-spoofing standards: SPF, DKIM, and DMARC

APPLICATION FOCUS

Use common productivity applications and communicate information clearly in academic and workplace settings.

Study prompt: Explain this idea in your own words and connect it to an example from your discipline.

Topic 5: Intended Learning Outcomes (ILOs)

Core Productivity & Communications Competencies

• • Engineer complex multi-section academic documents with dynamic numbering and automated tables of contents.

• • Design automated mail merge pipelines for dynamic reporting and student notifications.

• • Identify rows, columns, cells, and cell references in a worksheet.

• • Enter simple addition, subtraction, multiplication, and division formulas and use `SUM` or `AVERAGE`.

• • Organize small worksheets using clear headings, basic sorting, and simple charts.

• • Dissect Uniform Resource Locators (URLs) and trace HTTP/HTTPS client-server request-response cycles.

• • Contrast POP3 and IMAP4 mail retrieval protocols and formulate SPF/DKIM anti-spoofing policies.

Study prompt: Explain this idea in your own words and connect it to an example from your discipline.

Word Processing: Academic Document Engineering

Beyond Basic Typing: Structural Typography and Sectioning

• • Document Architecture: Hierarchical layers (Document → Sections → Paragraphs → Characters).

• • Section Breaks vs Page Breaks:

• - Page Break: Simply forces following text onto a new physical page; retains identical margins, orientation, and headers/footers.

• - Section Break (Next Page): Divides document into independent formatting zones with distinct margins, page numbering formats, and orientations (Portrait vs Landscape).

• • Thesis Compilation Rule at SUA:

• - Section 1 (Preliminary Pages): Roman numerals (i, ii, iii) centered at footer.

• - Section 2 (Chapter 1 onwards): Arabic numerals (1, 2, 3) aligned at top-right.

• - Unlink Headers/Footers: 'Link to Previous' MUST be disabled between sections.

Study prompt: Explain this idea in your own words and connect it to an example from your discipline.

Automated Academic Referencing & Tables of Contents

Leveraging Semantic Heading Styles for Document Integrity

• • The Semantic Heading Standard: Never manually format headings by changing font size and bolding.

• • Heading Styles (Heading 1, Heading 2, Heading 3):

• - Provides semantic structure recognized by document outline compilers.

• - Enables one-click generation of the automated Table of Contents (TOC).

• • Dynamic Updates: Modifying a chapter heading automatically updates the TOC and page numbers across a 200-page document via `F9` (Update Field).

• • Table of Figures & Captions: Figures and tables tagged with official captions (`Insert Caption`), generating automated lists of figures and tables.

• • Cross-Referencing: Dynamic links to numbered headings or figures that automatically track changes in figure numbering.

Study prompt: Explain this idea in your own words and connect it to an example from your discipline.

Mail Merge Architecture & Enterprise Dispatches

Mass Customization of Letters, Transcripts, and Advisory Notices

• • Core Architecture: Combines a standardized template with a structured transactional data source.

• • The Three Components of Mail Merge:

• 1. Main Document: Standardized text containing boilerplate prose and merge field placeholders (e.g., `«First_Name»`, `«Balance_Due»`).

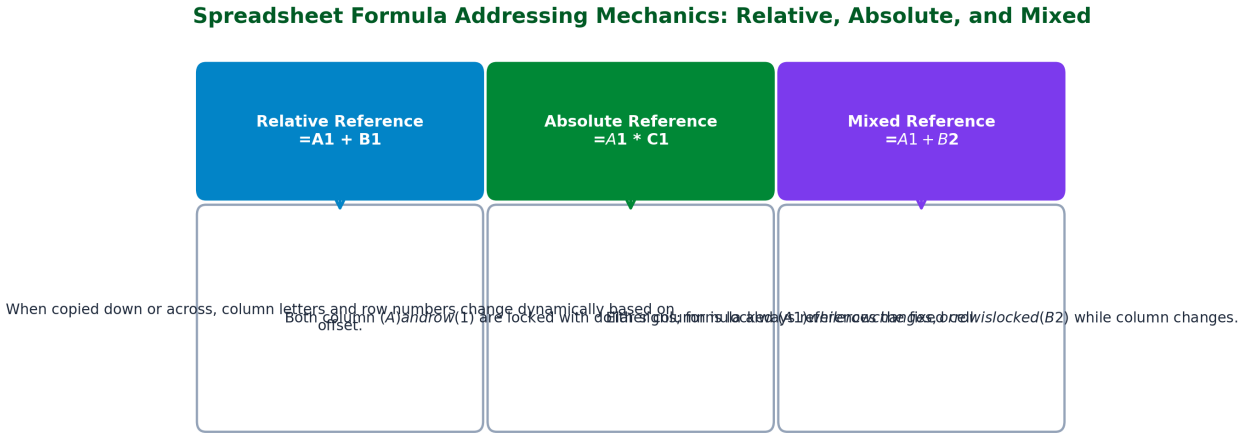
• 2. Data Source: Structured tabular dataset (Excel spreadsheet, Access database, CSV file) where columns represent fields and rows represent records.

• 3. Merged Output Document: Consolidated multi-page file or direct email dispatch containing individual customized records.

• • Conditional Logic Rules: Using Word `IF...THEN...ELSE` fields to dynamically insert customized advice based on numerical thresholds (e.g., 'Warning: Overdue Payment' if `Balance > 0`).

Study prompt: Explain this idea in your own words and connect it to an example from your discipline.

Spreadsheet Rows, Columns, and Cells



A worksheet is organized as columns, rows, and individual cells.

- • Columns are identified by letters and rows by numbers.
- • A cell reference combines its column and row, such as `B2`.
- • A formula can refer to values in other cells rather than repeating the values.
- • When a simple formula is copied to another row, its cell references normally adjust to that row.
- • Use clear headings so users can identify what each cell contains.

Study prompt: Explain this idea in your own words and connect it to an example from your discipline.

Simple Spreadsheet Formulas

A first calculation using farm data

• • A formula begins with an equals sign and uses cell references.

• • Addition: `=B2+C2`; subtraction: `=B2-C2`.

• • Multiplication: `=B2*C2`; division: `=B2/C2`.

• • Example: if B2 contains harvested kilograms and C2 contains the number of containers, `=B2/C2` gives the average per container.

• • Press Enter to display the result. Check that the selected cells contain the intended values.

Study prompt: Explain this idea in your own words and connect it to an example from your discipline.

Using SUM and AVERAGE

Built-in functions for introductory spreadsheet tasks

• • `=SUM(B2:B6)` adds the values in the selected range.

• • `=AVERAGE(B2:B6)` finds the mean of the values in the selected range.

• • Select the data range carefully and keep a clear label beside each result.

• • A spreadsheet updates a result when a value used by its formula changes.

• • Use these functions with small class examples such as crop yields, seedling counts, or animal feed records.

Study prompt: Explain this idea in your own words and connect it to an example from your discipline.

Checking and Presenting Spreadsheet Results

Simple review habits for class data

- • Confirm that headings identify the information in each column.
- • Use Sort to arrange records by a field such as date, crop, or animal ID.
- • Use a simple chart to show a small set of values clearly.
- • Check formulas and source data before sharing or printing a worksheet.
- • Do not use a chart to imply a conclusion that the data does not support.

Study prompt: Explain this idea in your own words and connect it to an example from your discipline.

Sorting and Charting Small Data Sets

Basic ways to organize and present worksheet information

- • Arrange related values in a labeled table, with one record on each row.
- • Use the Sort command to arrange records by a chosen column.
- • Use a basic chart when it helps communicate a small set of worksheet values.
- • Keep a copy of the original data before reorganizing a worksheet.
- • Explain what the chart shows and identify any limitations in the data.

Study prompt: Explain this idea in your own words and connect it to an example from your discipline.

The World Wide Web: Architecture & URLs

The Global Distributed Hypermedia System

• • Core Architecture: Client-server hypermedia infrastructure running over TCP/IP, proposed by Tim Berners-Lee (1989).

• • Anatomy of a Uniform Resource Locator (URL):

• `https://www.sua.ac.tz:443/academics/courses?dept=diit&level=100#syllabus`

• - Protocol / Scheme: `https://` (HyperText Transfer Protocol Secure).

• - Fully Qualified Domain Name (FQDN): `www.sua.ac.tz` (resolved via DNS to server IP).

• - Port Number: `:443` (standard encrypted TLS port; HTTP defaults to 80).

• - Path: `/academics/courses` (hierarchical resource location on web server).

• - Query String: `?dept=diit&level=100` (key-value parameters passed to backend script).

• - Fragment / Anchor: `#syllabus` (client-side jump target within web page).

Study prompt: Explain this idea in your own words and connect it to an example from your discipline.

The HTTP/HTTPS Request-Response Cycle

Stateless Communication and RESTful Operations

• • Client Request: Web browser opens TCP connection (port 80/443), sends HTTP request header:

• - HTTP Methods: `GET` (retrieve resource), `POST` (submit data/payload), `PUT` (replace resource), `DELETE` (remove resource).

• • Server Response: Server processes request and returns HTTP status code, response headers, and payload (HTML/JSON).

• • Standard HTTP Status Code Classes:

• - 1xx (Informational): Request received, continuing process.

• - 2xx (Success): `200 OK`, `201 Created`.

• - 3xx (Redirection): `301 Moved Permanently`, `302 Found`.

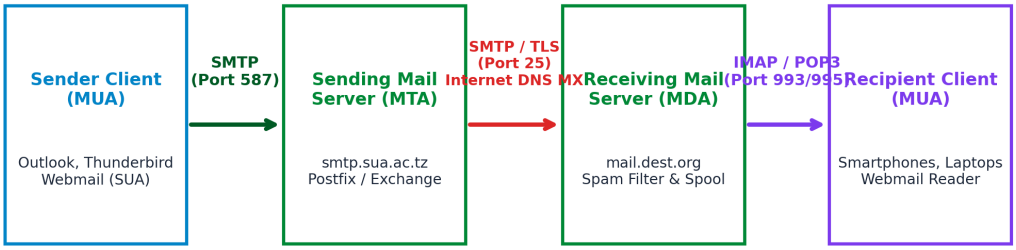
• - 4xx (Client Error): `400 Bad Request`, `401 Unauthorized`, `403 Forbidden`, `404 Not Found`.

• - 5xx (Server Error): `500 Internal Server Error`, `502 Bad Gateway`, `503 Service Unavailable`.

Study prompt: Explain this idea in your own words and connect it to an example from your discipline.

Electronic Mail Protocol Architecture

End-to-End Internet Electronic Mail (Email) Transport Architecture



The complete email pipeline: MUA, MSA, MTA, MDA, SMTP, POP3, and IMAP4.

• • Sender MUA (Outlook/Thunderbird) submits email to Mail Submission Agent (MSA) via SMTP (Port 587 with STARTTLS).

• • Mail Transfer Agent (MTA - Postfix/Sendmail) performs DNS MX record lookup and relays message across internet via SMTP (Port 25).

• • Receiving MTA transfers message to Mail Delivery Agent (MDA - Dovecot), storing it in the user's mailbox.

• • Recipient MUA retrieves or synchronizes mail via POP3 (Port 995) or IMAP4 (Port 993).

Study prompt: Explain this idea in your own words and connect it to an example from your discipline.

Email Retrieval: POP3 vs IMAP4

Store-and-Forward Download vs Real-Time Server Synchronization

• • Post Office Protocol version 3 (POP3 - RFC 1939):

• - Ports: Port 110 (plaintext), Port 995 (POP3S encrypted via TLS).

• - Operational Model: Connects, downloads all messages to local computer disk, and deletes them from server mailbox by default.

• - Limitations: Single-device model; read status, sent items, and custom folders do NOT synchronize across other devices (smartphone, office PC).

• • Internet Message Access Protocol version 4 (IMAP4 - RFC 3501):

• - Ports: Port 143 (STARTTLS), Port 993 (IMAPS encrypted via TLS).

• - Operational Model: All email messages, read/unread flags, drafts, and folder structures reside permanently on central server.

• - Advantages: Perfect multi-device synchronization; mandatory standard for modern enterprise email.

Study prompt: Explain this idea in your own words and connect it to an example from your discipline.

Email Anti-Spoofing Protocols: SPF, DKIM & DMARC

Combatting Domain Phishing and Fraudulent Impersonation

• • The Vulnerability: Standard SMTP headers allow any sender to forge the `From:` address arbitrarily.

• • 1. SPF (Sender Policy Framework - RFC 7208):

• - Domain owner publishes DNS TXT record listing authorized sending mail server IP addresses.

• - Example: `v=spf1 ip4:196.216.180.15 include:_spf.google.com -all`.

• • 2. DKIM (DomainKeys Identified Mail - RFC 6376):

• - Sending server cryptographically signs message headers with private key; recipient verifies signature using public key in sender's DNS record.

• • 3. DMARC (RFC 7489):

• - Coordinates SPF and DKIM alignment, instructing receiving servers what to do with failing messages (`none`, `quarantine`, `reject`) and generating forensic telemetry reports.

Study prompt: Explain this idea in your own words and connect it to an example from your discipline.

Academic Typography & Font Engineering

Serif vs Sans-Serif, Kerning, Tracking, and Readability

• • Typeface Categories:

• - Serif (e.g., Times New Roman, Georgia): Small decorative strokes at letter terminals; guides human eye across horizontal lines; traditional standard for printed body text.

• - Sans-Serif (e.g., Arial, Calibri, DejaVu Sans): Clean geometric glyphs without serifs; optimal legibility on digital computer monitors and slide presentations.

• - Monospaced (e.g., Courier New, Consolas): Every character occupies identical horizontal width; mandatory for code snippets, hexadecimal addresses, and tabular alignment.

• • Micro-Typography:

• - Leading (Line Spacing): Vertical distance between baselines (academic standard: 1.5 lines or 12 pt font with 18 pt leading).

• - Kerning: Dynamic spacing adjustment between specific character pairs (e.g., 'AV', 'To').

Study prompt: Explain this idea in your own words and connect it to an example from your discipline.

Document Security & The PDF/A Archival Standard

Digital Preservation, Password Protection & Immutability

• • Word Document Vulnerabilities: Editable `.docx` files risk accidental formula deletion, font substitution on different operating systems, and malicious macro execution.

• • Portable Document Format (PDF) by Adobe (ISO 32000): Preserves exact typographical layout, vector graphics, and embedded fonts across any operating system.

• • PDF/A Archival Standard (ISO 19005):

• - Specifically designed for 50+ year long-term digital preservation of SUA theses and university charters.

• - Strict Rules: Mandates 100% font embedding; forbids external font references, dynamic JavaScript, audio/video streams, and encryption.

• • Document Protection Controls: Setting Document Open Passwords (AES-128/256) and Permission Passwords restricting printing or text copying.

Study prompt: Explain this idea in your own words and connect it to an example from your discipline.

Entering and Checking Spreadsheet Data

Simple ways to reduce data-entry mistakes

- • Use consistent headings and units for each column.
- • Use a drop-down list when the class has a short set of approved labels, such as crop type.
- • Check spelling, missing values, and unexpected entries before using a worksheet.
- • Keep a protected copy of the original records before editing them.
- • Ask a lecturer or data owner when an entry is unclear; do not guess.

Study prompt: Explain this idea in your own words and connect it to an example from your discipline.

Formatting a Readable Worksheet

Presenting records clearly

- • Use a clear font and readable column widths.
- • Distinguish headings from data using consistent formatting.
- • Use borders or light shading sparingly to separate sections.
- • Keep colors high-contrast and do not rely on color alone to convey meaning.
- • Preview the worksheet before printing or sharing.

Study prompt: Explain this idea in your own words and connect it to an example from your discipline.

Reviewing and Sorting Worksheet Records

Keeping small class datasets understandable

- • Sort records by a meaningful field, such as date, animal ID, or crop name.
- • Use a filter to temporarily show records that match a label.
- • Confirm that each row stays together when sorting.
- • Clear filters before reporting the full dataset.
- • Save a separate copy before making substantial changes.

Study prompt: Explain this idea in your own words and connect it to an example from your discipline.

Dates and Text in a Worksheet

Entering information consistently without date calculations

- • Use one consistent date format throughout a worksheet.
- • Label text fields clearly, such as animal ID, crop, or location.
- • Avoid mixing dates, notes, and measurements in the same column.
- • Check that imported text has not been split or joined incorrectly.
- • Use simple headings so another student can understand the records.

Study prompt: Explain this idea in your own words and connect it to an example from your discipline.

Saving and Sharing a Spreadsheet

Protecting class and research records

- • Give the file a clear name that identifies the topic and date.
- • Save the worksheet in an approved location and make a backup copy.
- • Share only with people who are authorized to view the information.
- • Avoid putting personal or sensitive information in a public workbook.
- • Close or protect the workbook when using a shared computer.

Study prompt: Explain this idea in your own words and connect it to an example from your discipline.

Simple Charts from Worksheet Data

Choosing a clear visual for a small dataset

- • Select a small, labeled data range before inserting a chart.
- • Use a column chart to compare categories, such as crop types.
- • Use a line chart to show a trend across time when observations are ordered.
- • Add a title and axis labels so the chart can be understood on its own.
- • Check that the selected chart type does not hide important information.

Study prompt: Explain this idea in your own words and connect it to an example from your discipline.

Web Browser Engine Architecture

Parsing HTML, Constructing the DOM, and JavaScript Execution

• • Core Subsystems of a Modern Web Browser:

• 1. User Interface: Address bar, back/forward buttons, bookmarking menu.

• 2. Browser Engine: Coordinates actions between UI and rendering engine.

• 3. Rendering Engine: Parses HTML and CSS to render graphical web pages (Blink in Chrome/Edge, Gecko in Firefox, WebKit in Safari).

• 4. JavaScript Engine: Compiles and executes script code (V8 in Chrome, SpiderMonkey in Firefox).

• 5. Networking Layer: Handles HTTP/HTTPS, DNS resolution, and TLS handshakes.

• • Critical Rendering Path:

• - Parses HTML → Builds Document Object Model (DOM) tree.

• - Parses CSS → Builds CSS Object Model (CSSOM) tree.

• - Combines DOM + CSSOM → Render Tree → Layout phase → Paint phase.

Study prompt: Explain this idea in your own words and connect it to an example from your discipline.

Web State Management: Cookies vs LocalStorage

Maintaining Stateful Sessions Across Stateless HTTP

• • HTTP is Stateless: Each HTTP request is completely independent; servers do not remember previous requests without state mechanisms.

• • HTTP Cookies (RFC 6265): Small key-value text pairs (max 4 KB) stored by browser and transmitted back to server with EVERY HTTP request header.

• - Session Cookies: Stored in volatile RAM; deleted when browser closes.

• - Persistent Cookies: Stored on disk until expiration date (e.g., 'Remember Me' logins).

• - Security Flags: `Secure` (transmitted only over HTTPS), `HttpOnly` (inaccessible to JavaScript, neutralizing XSS credential theft), `SameSite=Strict` (prevents CSRF attacks).

• • HTML5 Web Storage: `localStorage` (persists indefinitely, 5–10 MB, client-side only) and `sessionStorage` (cleared on tab close).

Study prompt: Explain this idea in your own words and connect it to an example from your discipline.

Domain Name System (DNS) Hierarchical Resolution

Translating Human Hostnames into Machine IP Addresses

• • The Phonebook of the Internet: Resolves `www.sua.ac.tz` to IPv4 `196.216.180.15`.

• • The Hierarchical Tree Architecture:

• 1. Root DNS Servers (`.`): 13 logical root server clusters managed globally via Anycast routing.

• 2. Top-Level Domain (TLD) Servers: Manage gTLDs (`.com`, `.org`) and ccTLDs (`.tz`, `.ke`).

• 3. Authoritative Nameservers: Hold official master DNS records for the organization domain (`sua.ac.tz`).

• • Essential DNS Record Types:

• - `A` Record: Maps hostname to 32-bit IPv4 address.

• - `AAAA` Record: Maps hostname to 128-bit IPv6 address.

• - `CNAME` (Canonical Name): Alias mapping one hostname to another.

• - `MX` (Mail Exchanger): Identifies mail servers responsible for domain email.

• - `TXT`: Arbitrary text; houses SPF, DKIM public keys, and domain ownership tokens.

Study prompt: Explain this idea in your own words and connect it to an example from your discipline.

Transport Layer Security (TLS 1.3) Handshake

Establishing Encrypted, Tamper-Proof HTTPS Sessions

• • Core Goals: Confidentiality (AES-GCM encryption), Integrity (HMAC authentication), Server Authenticity (X.509 digital certificate).

• • TLS 1.3 Rapid Handshake Sequence (1-RTT Round Trip Time):

• 1. `ClientHello`: Client sends supported cipher suites and ephemeral Diffie-Hellman public key share.

• 2. `ServerHello`: Server selects cipher suite, sends its own key share, its X.509 certificate, and digital signature.

• 3. Session Key Derivation: Both parties use the key-exchange process to establish a shared session key.

• 4. Encrypted Transmission: All subsequent HTTP requests and responses encrypted with high-speed AES-256-GCM.

• • Perfect Forward Secrecy (PFS): Even if the server's master private key is stolen years later, past recorded encrypted sessions cannot be decrypted.

Study prompt: Explain this idea in your own words and connect it to an example from your discipline.

Cloud Productivity Ecosystems in Academic Research

Microsoft 365, Google Workspace, and Private Clouds

• • Collaborative Cloud Architecture: Real-time co-authoring using Operational Transformation (OT) and Conflict-Free Replicated Data Types (CRDT).

• • Enterprise Cloud Suites:

• - Microsoft 365: Word Online, Excel Online, OneDrive, SharePoint; deep enterprise Active Directory integration.

• - Google Workspace: Google Docs, Sheets, Drive; streamlined browser-native collaboration.

• - Self-Hosted Private Clouds (Nextcloud / ownCloud): Deployed on SUA DIIT servers to ensure sensitive national agricultural research data remains sovereign within Tanzanian borders.

• • Cloud Storage Mechanics: Delta sync (transmitting only modified byte blocks of large files) and continuous versioning allowing rollback to previous revisions.

Study prompt: Explain this idea in your own words and connect it to an example from your discipline.

Topic 5: Key Takeaways & Synthesis Matrix

Summary of Applications & Networking for Examination Mastery

• • Document Engineering: Section breaks isolate independent formatting; heading styles empower dynamic tables of contents; mail merge automates mass communications.

• • Spreadsheet Referencing: Relative (`A1`) shifts dynamically; Absolute (`$A$1`) stays locked; Mixed (`$A1`/`A$1`) locks one coordinate.

• • Simple Spreadsheet Work: Cell references, basic arithmetic formulas, `SUM`, and `AVERAGE` support introductory class tasks.

• • Web Infrastructure: URLs specify protocol, host, port, path, and query; HTTP request-response cycle handles stateless web interactions.

• • Email Architecture: SMTP relays messages server-to-server; IMAP4 provides multi-device synchronized retrieval; SPF/DKIM/DMARC eliminate domain spoofing.

Study prompt: Explain this idea in your own words and connect it to an example from your discipline.

Review Quiz & Knowledge Check

Self-Testing Key Software & Protocol Principles

• 1. Why is a Section Break mandatory if you need Chapter 4 of a thesis in Landscape and Chapter 5 in Portrait?

• 2. When copying formula `=$C$2 · D5` from cell E5 to E6, what will the resulting formula be? (Ans: `=$C$2 · D6`).

• 3. Name a suitable chart for comparing crop types and explain how you would label it.

• 4. Differentiate between POP3 and IMAP4 regarding multi-device message synchronization.

• 5. What does the `-all` mechanism at the end of an SPF DNS TXT record instruct receiving mail servers to do?

Study prompt: Explain this idea in your own words and connect it to an example from your discipline.

Reading Assignments & Homework Notice

Preparation for Next Lecture & Continuous Assessment

• • Required Readings:

• - Microsoft Support. Overview of formulas in Excel. https://support.microsoft.com/en-us/excel/get-started/overview-of-formulas-in-excel

• - Smith, J. L., Wang, S., & Mortati, J. - Information Systems for Business and Beyond (2019), Chapter 5: Networking and Communication. https://opentextbook.site/informationsystems2019/

• • Homework Assignment 5 Notice:

• - Available on SUA Moodle / Assignment directory: `BIT_111_Assignment_5.pdf`.

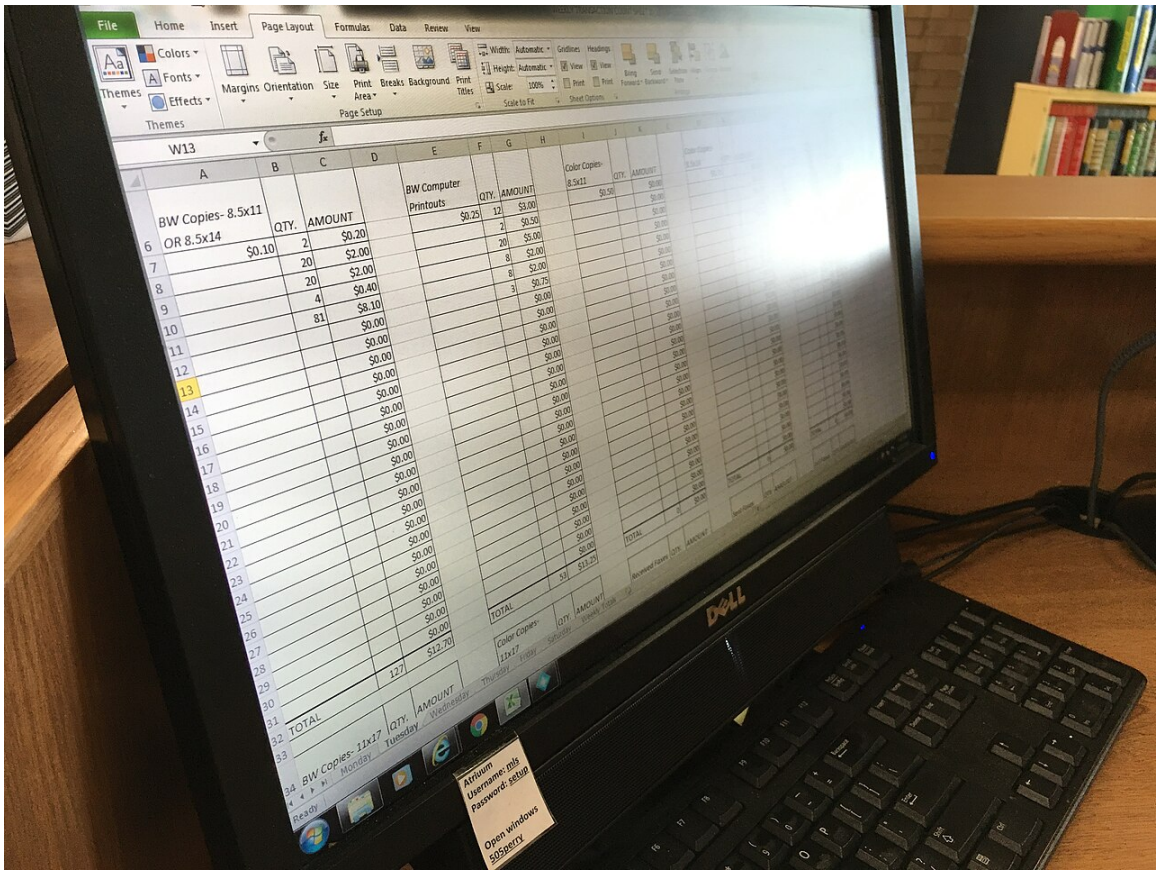
• - Total Allocation: 100 Marks (Covers document structuring, simple spreadsheet use, and email protocols).

• - Submission Deadline: End of Week 10 by 17:00 EAT.

• • Next Lecture Preview: Topic 6 — Data Security and Malware Defense (Physical Protection, Cryptography, Anti-Malware).

Study prompt: Explain this idea in your own words and connect it to an example from your discipline.

ONLINE FIGURE 1: Closeup of Excel Spreadsheet template to track printouts (29911005444).jpg

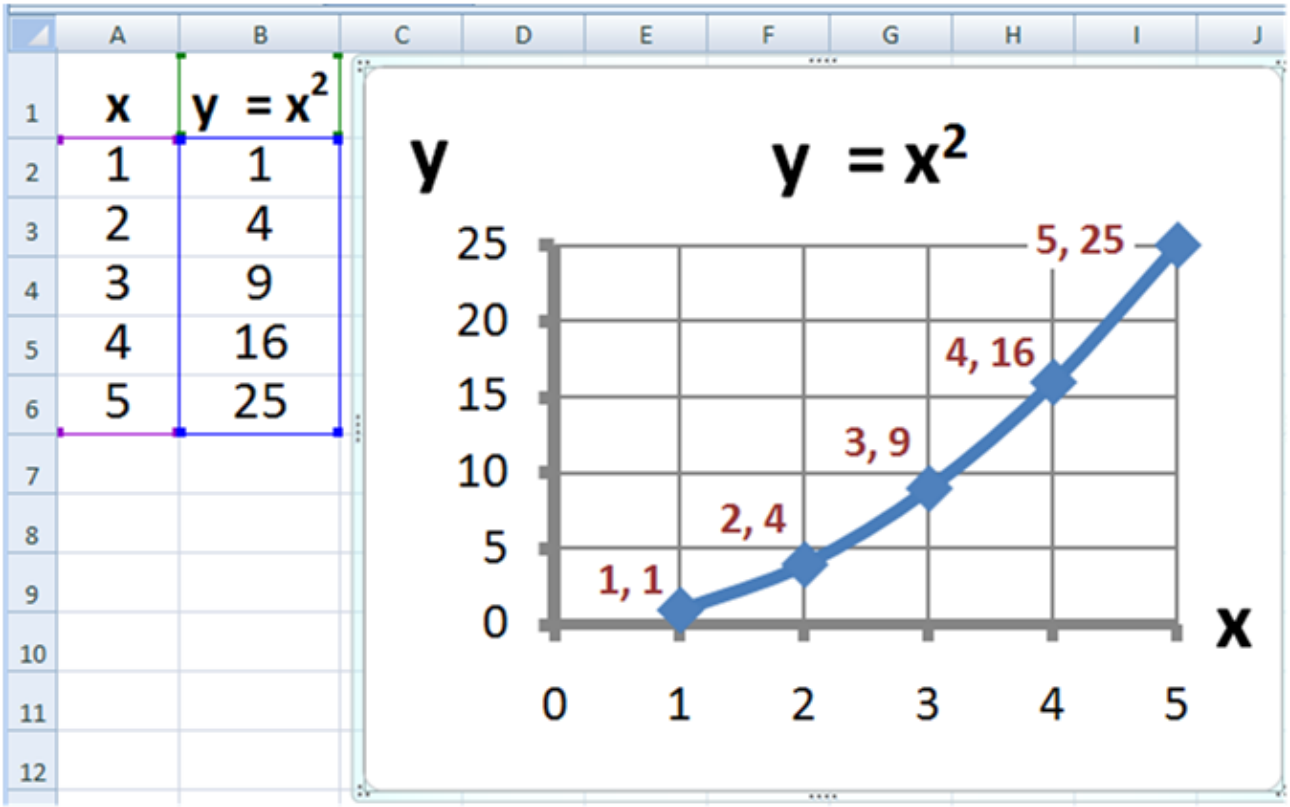


What to notice: Closeup of Excel Spreadsheet template to track printouts (29911005444)

Topic 5 connection: Microcomputer Applications. Discuss what the figure helps you recognize and what information it does not provide by itself.

Attribution: Texas State Library and Archives Commission from Austin, TX, United States. Wikimedia Commons source page. License: CC BY 2.0.

ONLINE FIGURE 2: Excel chart.PNG

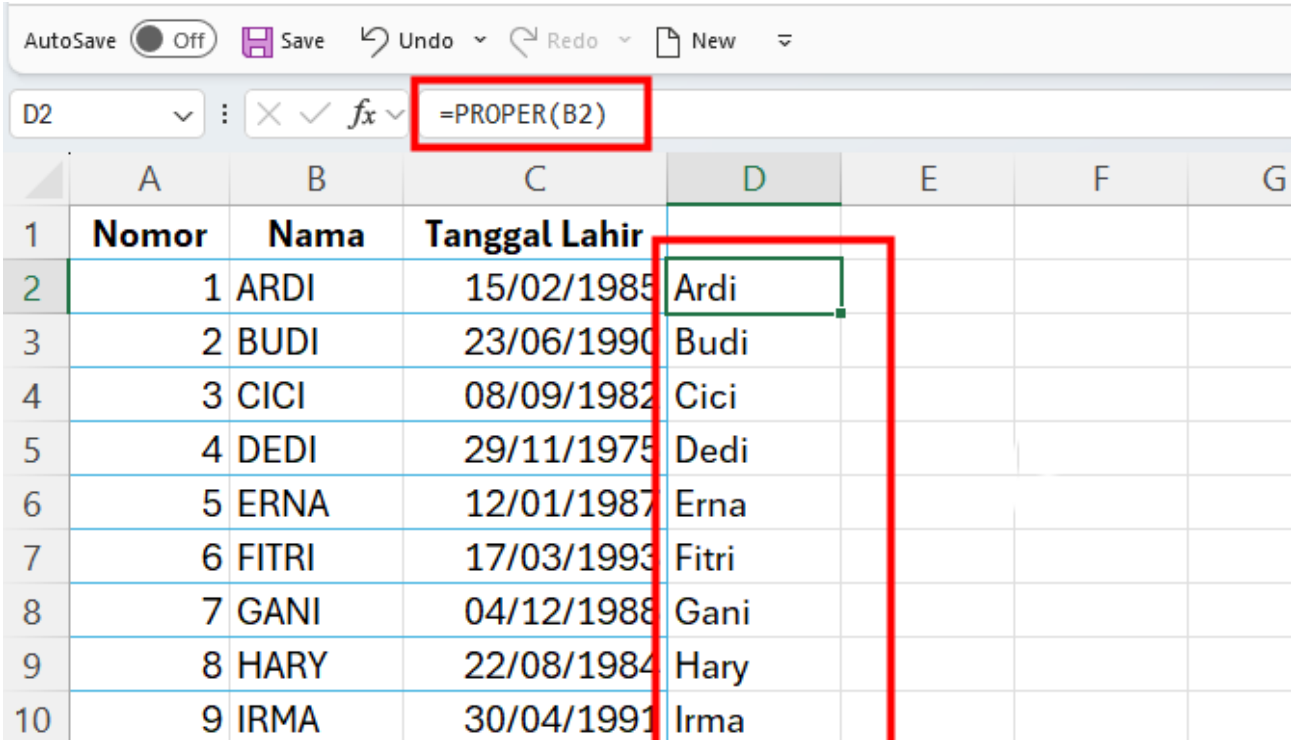


What to notice: Excel chart

Topic 5 connection: Microcomputer Applications. Discuss what the figure helps you recognize and what information it does not provide by itself.

Attribution: Brews ohare. Wikimedia Commons source page. License: Public domain.

ONLINE FIGURE 3: Fungsi Proper Pada Microsoft Excel.png

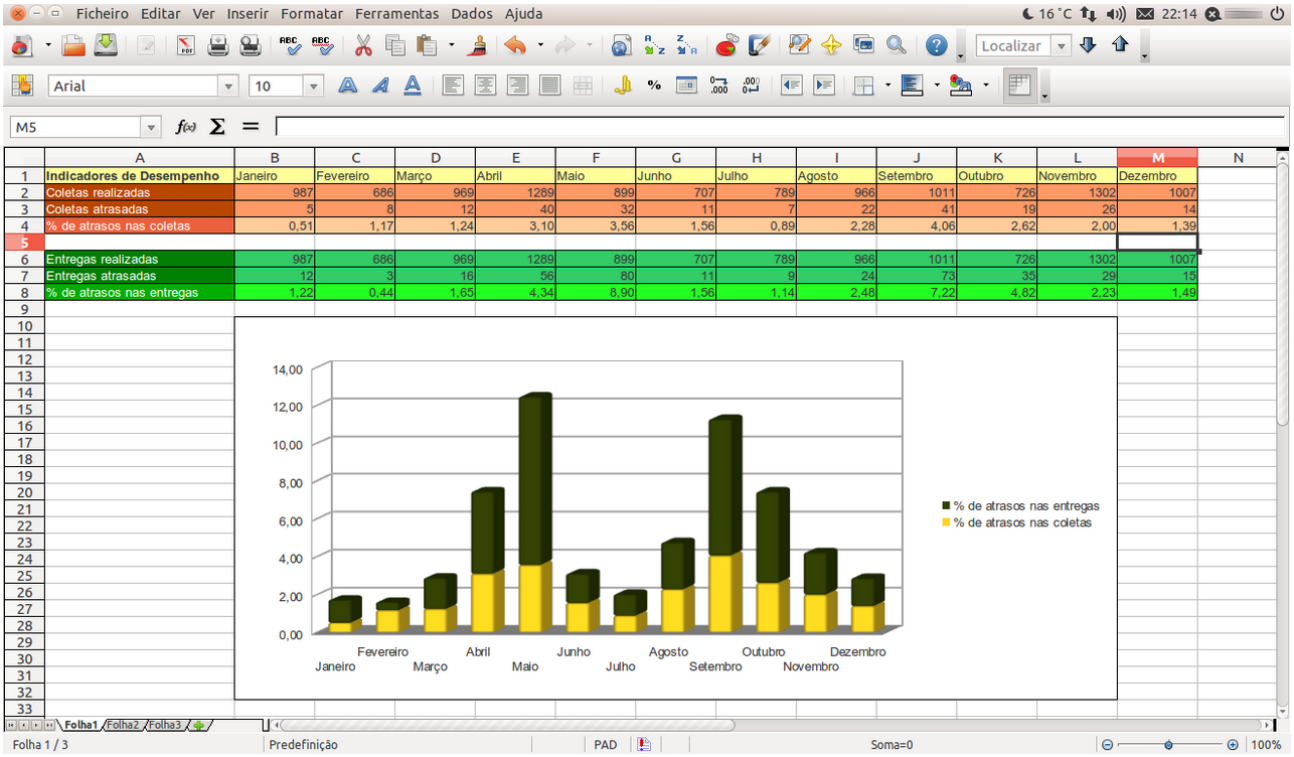


What to notice: Fungsi Proper Pada Microsoft Excel

Topic 5 connection: Microcomputer Applications. Discuss what the figure helps you recognize and what information it does not provide by itself.

Attribution: Umulhidayah. Wikimedia Commons source page. License: CC BY-SA 4.0.

ONLINE FIGURE 4: Libreoffice-calc.png

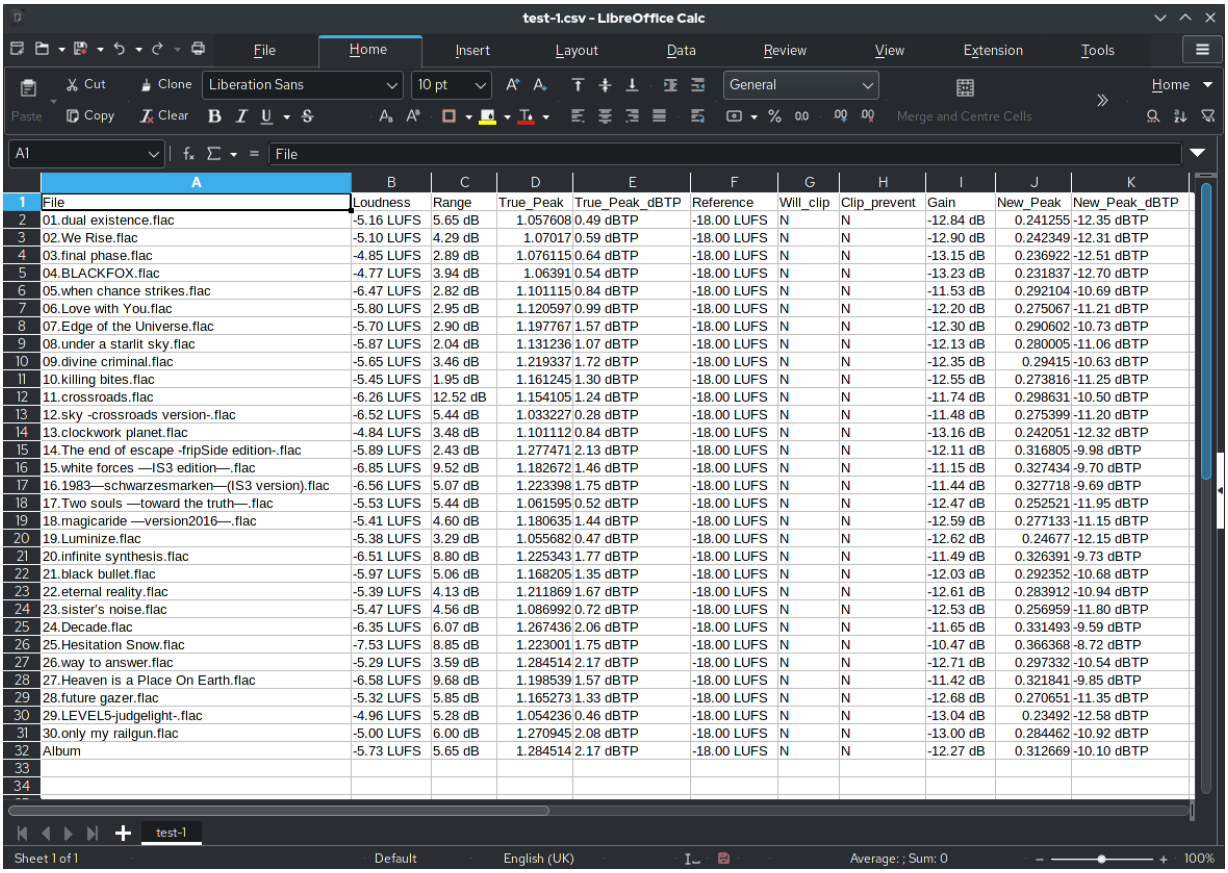


What to notice: Libreoffice-calc

Topic 5 connection: Microcomputer Applications. Discuss what the figure helps you recognize and what information it does not provide by itself.

Attribution: The Document Foundation and Canonical. Wikimedia Commons source page. License: CC0.

ONLINE FIGURE 5: LibreOffice 7.2.4.1 Calc with csv screenshot.png



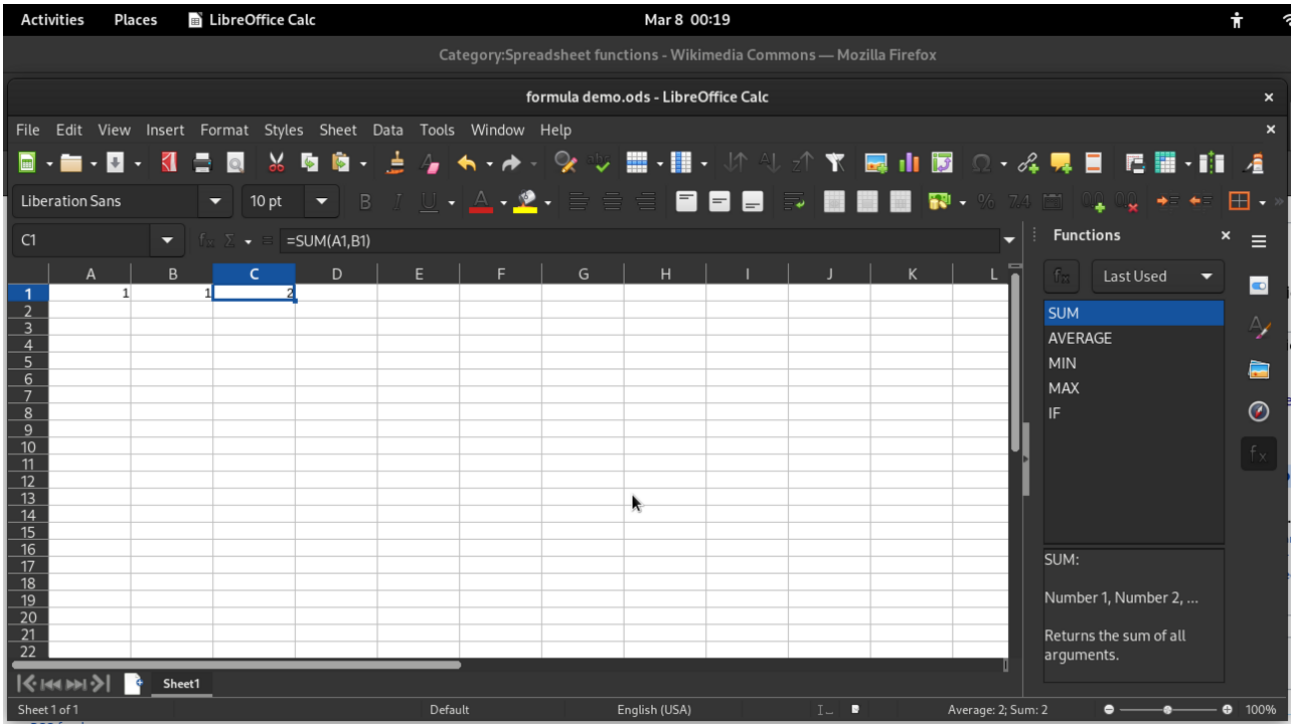
	A	B	C	D	E	F	G	H	I	J	K
1	File	Loudness	Range	True_Peak	True_Peak_dBTP	Reference	Will_clip	Clip_prevent	Gain	New_Peak	New_Peak_dBTP
2	01.dual existence.flac	-5.16 LUFS	5.65 dB	1.057608	0.49 dBTP	-18.00 LUFS	N	N	-12.84 dB	0.241255	-12.35 dBTP
3	02.We Rise.flac	-5.10 LUFS	4.29 dB	1.07017	0.59 dBTP	-18.00 LUFS	N	N	-12.90 dB	0.242349	-12.31 dBTP
4	03.final phase.flac	-4.85 LUFS	2.89 dB	1.076115	0.64 dBTP	-18.00 LUFS	N	N	-13.15 dB	0.236922	-12.51 dBTP
5	04.BLACKFOX.flac	-4.77 LUFS	3.94 dB	1.06391	0.54 dBTP	-18.00 LUFS	N	N	-13.23 dB	0.231837	-12.70 dBTP
6	05.when chance strikes.flac	-6.47 LUFS	2.82 dB	1.101115	0.84 dBTP	-18.00 LUFS	N	N	-11.53 dB	0.292104	-10.69 dBTP
7	06.Love with You.flac	-5.80 LUFS	2.95 dB	1.120597	0.99 dBTP	-18.00 LUFS	N	N	-12.20 dB	0.275067	-11.21 dBTP
8	07.Edge of the Universe.flac	-5.70 LUFS	2.90 dB	1.197767	1.57 dBTP	-18.00 LUFS	N	N	-12.30 dB	0.290602	-10.73 dBTP
9	08.under a starlit sky.flac	-5.87 LUFS	2.04 dB	1.131236	1.07 dBTP	-18.00 LUFS	N	N	-12.13 dB	0.280005	-11.06 dBTP
10	09.divine criminal.flac	-5.65 LUFS	3.46 dB	1.219337	1.72 dBTP	-18.00 LUFS	N	N	-12.35 dB	0.29415	-10.63 dBTP
11	10.killing bites.flac	-5.45 LUFS	1.95 dB	1.161245	1.30 dBTP	-18.00 LUFS	N	N	-12.55 dB	0.273816	-11.25 dBTP
12	11.crossroads.flac	-6.26 LUFS	12.52 dB	1.154105	1.24 dBTP	-18.00 LUFS	N	N	-11.74 dB	0.298631	-10.50 dBTP
13	12.sky -crossroads version-.flac	-6.52 LUFS	5.44 dB	1.033227	0.28 dBTP	-18.00 LUFS	N	N	-11.48 dB	0.275399	-11.20 dBTP
14	13.clockwork planet.flac	-4.84 LUFS	3.48 dB	1.101112	0.84 dBTP	-18.00 LUFS	N	N	-13.16 dB	0.242051	-12.32 dBTP
15	14.The end of escape -fripSide edition-.flac	-5.89 LUFS	2.43 dB	1.277471	2.13 dBTP	-18.00 LUFS	N	N	-12.11 dB	0.316805	-9.98 dBTP
16	15.white forces —IS3 edition—.flac	-6.85 LUFS	9.52 dB	1.182672	1.46 dBTP	-18.00 LUFS	N	N	-11.15 dB	0.327434	-9.70 dBTP
17	16.1983—schwarzesmarken—(IS3 version).flac	-6.56 LUFS	5.07 dB	1.223398	1.75 dBTP	-18.00 LUFS	N	N	-11.44 dB	0.327718	-9.69 dBTP
18	17.Two souls —toward the truth—.flac	-5.53 LUFS	5.44 dB	1.061595	0.52 dBTP	-18.00 LUFS	N	N	-12.47 dB	0.252521	-11.95 dBTP
19	18.magicaride —version2016—.flac	-5.41 LUFS	4.60 dB	1.180635	1.44 dBTP	-18.00 LUFS	N	N	-12.59 dB	0.277133	-11.15 dBTP
20	19.Luminize.flac	-5.38 LUFS	3.29 dB	1.055682	0.47 dBTP	-18.00 LUFS	N	N	-12.62 dB	0.24677	-12.15 dBTP
21	20.infinite synthesis.flac	-6.51 LUFS	8.80 dB	1.225343	1.77 dBTP	-18.00 LUFS	N	N	-11.49 dB	0.326391	-9.73 dBTP
22	21.black bullet.flac	-5.97 LUFS	5.06 dB	1.168205	1.35 dBTP	-18.00 LUFS	N	N	-12.03 dB	0.292352	-10.68 dBTP
23	22.eternal reality.flac	-5.39 LUFS	4.13 dB	1.211869	1.67 dBTP	-18.00 LUFS	N	N	-12.61 dB	0.283912	-10.94 dBTP
24	23.sister's noise.flac	-5.47 LUFS	4.56 dB	1.086992	0.72 dBTP	-18.00 LUFS	N	N	-12.53 dB	0.256959	-11.80 dBTP
25	24.Decade.flac	-6.35 LUFS	6.07 dB	1.267436	2.06 dBTP	-18.00 LUFS	N	N	-11.65 dB	0.331493	-9.59 dBTP
26	25.Hesitation Snow.flac	-7.53 LUFS	8.85 dB	1.223001	1.75 dBTP	-18.00 LUFS	N	N	-10.47 dB	0.366368	-8.72 dBTP
27	26.way to answer.flac	-5.29 LUFS	3.59 dB	1.284514	2.17 dBTP	-18.00 LUFS	N	N	-12.71 dB	0.297332	-10.54 dBTP
28	27.Heaven is a Place On Earth.flac	-6.58 LUFS	9.68 dB	1.198539	1.57 dBTP	-18.00 LUFS	N	N	-11.42 dB	0.321841	-9.85 dBTP
29	28.future gazer.flac	-5.32 LUFS	5.85 dB	1.165273	1.33 dBTP	-18.00 LUFS	N	N	-12.68 dB	0.270651	-11.35 dBTP
30	29.LEVEL5-judgelight-.flac	-4.96 LUFS	5.28 dB	1.054236	0.46 dBTP	-18.00 LUFS	N	N	-13.04 dB	0.23492	-12.58 dBTP
31	30.only my railgun.flac	-5.00 LUFS	6.00 dB	1.270945	2.08 dBTP	-18.00 LUFS	N	N	-13.00 dB	0.284462	-10.92 dBTP
32	Album	-5.73 LUFS	5.65 dB	1.284514	2.17 dBTP	-18.00 LUFS	N	N	-12.27 dB	0.312669	-10.10 dBTP
33											
34											

What to notice: LibreOffice 7.2.4.1 Calc with csv screenshot

Topic 5 connection: Microcomputer Applications. Discuss what the figure helps you recognize and what information it does not provide by itself.

Attribution: The Document Foundation, Alessandro Ghedini, Matthias C. Hormann, and contributors; screenshot: VulcanSphere. Wikimedia Commons source page. License: MPL 2.

ONLINE FIGURE 6: SUM formula in LibreOffice Calc.png

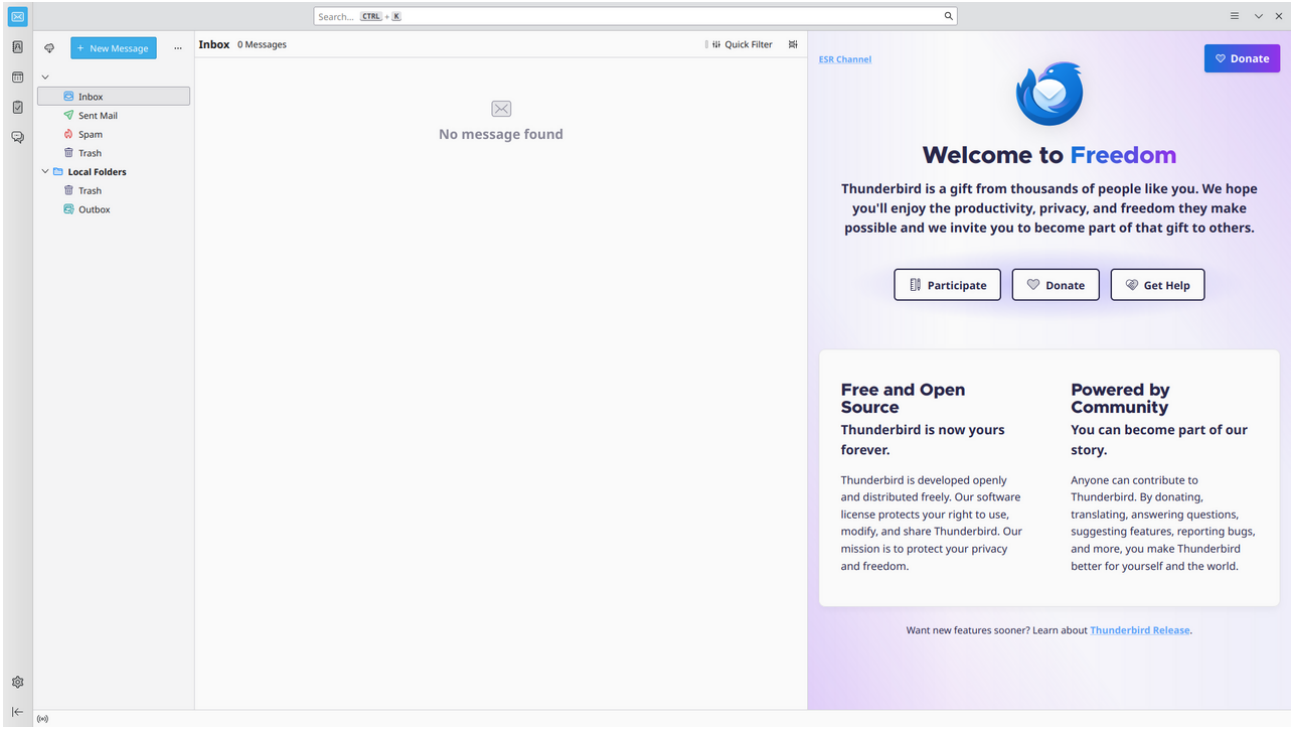


What to notice: SUM formula in LibreOffice Calc

Topic 5 connection: Microcomputer Applications. Discuss what the figure helps you recognize and what information it does not provide by itself.

Attribution: Arlo Barnes. Wikimedia Commons source page. License: CC0.

ONLINE FIGURE 7: Mozilla Thunderbird 140 ESR on Debian 13 (Trixie) and KDE Plasma 6.png



What to notice: Mozilla Thunderbird 140 ESR on Debian 13 (Trixie) and KDE Plasma 6

Topic 5 connection: Microcomputer Applications. Discuss what the figure helps you recognize and what information it does not provide by itself.

Attribution: Victor Matheus Amaral. Wikimedia Commons source page. License: CC0.

ONLINE FIGURE 8: CPT 8100 Word Processor Desktop Microcomputer 5185A65A.png



What to notice: CPT 8100 Word Processor Desktop Microcomputer 5185A65A

Topic 5 connection: Microcomputer Applications. Discuss what the figure helps you recognize and what information it does not provide by itself.

Attribution: LehmanUMN. Wikimedia Commons source page. License: CC0.

READING AND IMAGE CREDITS

Smith, J. L., Wang, S., & Mortati, J. (2019). Information Systems for Business and Beyond. Open textbook. https://opentextbook.site/informationsystems2019/

Vermaat, M. E., Sebok, S. L., Freund, S. M., Campbell, J. T., & Frydenberg, M. (2018). Discovering Computers: Digital Technology, Data, and Devices (16th ed.). Cengage. https://faculty.cengage.com/works/9781337285100

Microsoft. (n.d.). Overview of formulas in Excel. Microsoft Support. https://support.microsoft.com/en-us/excel/get-started/overview-of-formulas-in-excel

Online image credits appear beside each image. Images were selected from Wikimedia Commons and included only where a public-domain or stated reuse license could be checked on the file page.